

An LLM Agent for the Selection, Validation and Interpretation of Statistical Methods

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40 questions a professor is likely to ask, grouped by topic, each with an answer in my own words. The last section is how I would take the project further.

Why this problem

Why did you choose this problem?

Because AI tools are being used more and more to analyse data, often by people who aren't statisticians. And the mistakes they make aren't the obvious kind. The numbers add up and the report reads well, but the method can still be wrong for the data. That kind of mistake is hard to spot, so I wanted to see if it could be prevented.

Why does it matter if an AI uses the wrong statistical test?

Because the conclusion can be wrong even when everything looks right. If you use a test that assumes independent observations on data that isn't independent, you can end up much more confident than you should be. Someone could make a business or policy decision on that, and nothing in the report would warn them.

Why can't the AI just work out the study design from the data?

Because the information isn't there. A spreadsheet doesn't tell you whether the same person appears twice, or whether the rows are in time order, or whether measurements were grouped. Those are facts about how the data was collected. That's why I give the system a design card with that information, and score whether it actually uses it.

Who would actually use something like this?

Anyone who needs to analyse data but can't check the method themselves. Business analysts, researchers outside statistics, policy teams, students. They're exactly the people most likely to trust an AI's answer, and the least able to catch it when the method is wrong.

How the system is designed

Why did you limit it to 14 methods?

Mainly so I could score it. If the AI could use any method it liked, there'd be no clear right answer to mark it against, and it would turn into grading essays. Fourteen methods covers the most common kinds of question, like comparing groups, testing association and regression, which is enough to test the idea properly.

Why didn't you let the AI write and run its own code?

Because it could then skip the procedure completely. It could just write whatever analysis it wanted, and the three systems wouldn't be doing comparable work anymore. Keeping the tools fixed means the only thing that differs between them is the procedure.

Why is declining counted as a correct answer?

Because sometimes it genuinely is the right answer. If every method you have is invalid for the data, the responsible thing is to say so. In my benchmark, 12 of the 64 cases are like that. If I didn't reward declining, I'd be rewarding the system for answering things it shouldn't have answered.

What exactly is a design card, and who fills it in?

It's a short description of how the study was set up. Whether the observations are independent, whether anything is paired, whether there's a time order, whether data is grouped. In my benchmark it's filled in as part of each case. In a real tool, that's something the system would need to ask the user, and that's one of the things I'd build next.

How does the provenance contract actually work?

Every real calculation stores its result under a label. The report format refuses any number typed straight into the text, so the model has to write a reference to the label instead. At the end, each reference is swapped for the stored value. If it references something that was never calculated, the whole run fails. So a made-up number can't get into the report at all.

If made-up numbers are impossible in system C, isn't its accuracy on numbers automatic?

Yes, and that's why I don't count it as a result. It's guaranteed by the way it's built. What I actually measure is how often the model tried to reference something that didn't exist, and that's reported separately.

Why did you build three systems instead of just two?

Because two wouldn't tell me what's causing the improvement. If I only compared a plain AI with my full system, I couldn't tell whether the gain came from the tools or from the procedure. Adding B, which has the tools but no procedure, lets me separate those two things.

How did you make sure the comparison was fair?

All three systems share the same core code. The same tools, the same result store, the same report format. The only thing that changes is the order they work in. So there's no place in the code where one of the baselines could have been given something worse.

The benchmark

How did you build the 64 cases?

Each case is generated from a definition rather than written by hand. 48 are synthetic, where I control how the data is generated, and 16 come from four public datasets. Each case comes with the data, the question, the design card and the correct answer.

How do you know your answer key is right?

For 48 of the 64 cases, the correct answer comes from how the data was built, not from my judgement. And I base it on the data that actually came out, not what I meant to generate. For the 16 public cases, I checked every label against the real data. That found one label that was wrong, and I fixed it.

Why use synthetic data? Isn't real data better?

Real data is more realistic, but with synthetic data I know the true answer for certain, because I know exactly how it was made. That makes the scoring trustworthy. I used both, so the synthetic cases give me a reliable answer key and the real datasets check it holds up on messy data.

Why did you choose these four datasets?

Between them they cover the different kinds of question in the library. Bank Marketing and Online Shoppers give categorical and logistic problems, Student Performance gives group comparisons and regression, and Seoul Bike gives the design-hazard cases where the rows aren't independent.

The evaluation and the statistics

Why did you keep 48 cases hidden?

To avoid tuning the system to the test. If I'd adjusted the instructions while looking at the same cases I measured on, a good score would just mean I'd fitted it to those answers. Keeping 48 cases locked away until the end means the result reflects cases the system had never been shaped around.

When did you freeze everything?

Before the held-out run. I fixed the instructions given to the model, the model version and the code, and recorded a fingerprint of the instructions. So nothing could change between the development stage and the final test.

Why McNemar's test?

Because every case was answered by all three systems, so the results are paired. And each result is just right or wrong. McNemar's test is built for exactly that. It only looks at the cases where the two systems disagreed, which is where the real difference shows up.

Why did you compare case by case using a majority?

Each case was run twice, and McNemar's test assumes each pair is independent. Two runs of the same case aren't two separate cases, so counting both would make the result look stronger than it is. So I take one answer per case, the majority across the runs.

Why only two runs per case?

It came down to resources. Two runs were enough to show the main result and to see how consistent the systems were, but not enough to fully describe the variation. More runs is one of the first things I'd add.

Why did you use Claude Haiku 4.5 and not a bigger model?

I'm only claiming results for Haiku 4.5, because that's what I tested. The system itself works with any model, and testing on bigger ones is a clear next step. Whether the procedure still helps as models get stronger is actually one of the more interesting open questions.

Why didn't you set the temperature to zero?

On current models that setting has been removed, and it returns an error if you send it. So instead I fixed the model version and the effort level, froze the instructions, and measured how much the results varied between runs, rather than assuming they wouldn't.

What do the 95% intervals on your charts mean?

They show the range the true accuracy is likely to fall in, given how many cases I tested. A wide interval means I'm less certain. I used Wilson intervals because they behave properly near 0% and 100%, which matters for results like the abstention rates.

The results

What is the most important result?

That system C beat system B by 25 percentage points, with a p-value of 0.0018. B has exactly the same tools as C, so that difference comes from the procedure. My target at the start was 10 points.

Why did B do worse than A, even though B has tools?

Because having tools let B produce an answer for everything, including the cases where it should have stopped. B never declined once on the cases where every method was invalid. So the tools made it more confident, not more careful. That's really the point of the project.

C over A isn't significant. So is the procedure really better than nothing?

I can't claim that yet. It's 12.5 points in C's favour, and the cases where they disagreed split seven to one for C. But with 48 cases it isn't enough to rule out chance. When I worked out the power afterwards, my test only had about a 45% chance of detecting a difference that size. So it's unresolved, not a negative result.

Why is B at 100% on grounding but the worst on accuracy?

They measure different things. Grounding asks whether the numbers in the report came from real calculations. B uses real tools, so its numbers are real. Accuracy asks whether it picked the right method, and that's where B goes wrong. It calculates the wrong thing correctly.

Doesn't C just look better because it refuses to answer more?

No. C never declined a case that actually had a valid method. That rate was zero. Every time it declined, declining was the right call. And on the cases that could be answered, it picked the right method 91% of the time, which is higher than both A and B.

The expert policy scored 100%. Doesn't that prove it works?

No, that score is circular. The expert policy is the rule set that produced the answer key, so of course it gets everything right. I only use it to check that my testing setup works. The more useful result is a deliberately weak policy inside the same structure. It only got 60% and never declined, which shows the structure on its own isn't enough.

Difficulties and limitations

What was the hardest part?

Honestly, checking my own measurements. The system worked, but two things in how I was measuring it were wrong, and neither one showed up as an error. One would have made system A look 6 points better, and the other made the same data give different p-values each time I ran it. I only found them by going through the results carefully.

Why did you keep the runs that ended in errors?

Because none of them were technical failures. Each one had already chosen a method before its report was rejected, and 37 of them had chosen correctly. If I'd thrown them out, I'd have hidden the cases where system A was worst, and it would have looked 6 points better than it is.

What went wrong with the interpretation scoring?

I had two people score the written explanations blind, but they didn't agree with each other. The agreement score was minus 0.044, which is worse than chance. The problem was the rubric. One of its levels could be read two ways. I rebuilt it, and left that measure out rather than report a number I couldn't trust.

When does your system still fail?

It's not perfect. On the design-hazard cases it missed 4 of 18 and went ahead with a method anyway. On the answerable cases it picked the wrong method about 9% of the time, mostly from diagnostic reasoning rather than the design check. And 8 of its reports were rejected for referencing something that hadn't been calculated.

When shouldn't someone use this?

For any design the library doesn't cover, like repeated measures, mixed models or time series. In those cases it's built to decline, which is safe, but it can't give you an analysis. It also depends on the design card being filled in honestly. If the design information is wrong, its decision will be too.

Challenging questions

Isn't this just prompt engineering?

It's more than wording. The order is enforced in code, so the model can't skip the design check, and the report format physically rejects typed numbers. That's structure, not phrasing. But I'd be honest that structure alone isn't enough. A weak policy inside the same structure only reached 60%. What works is the structure plus good judgement at each step.

Couldn't system B just do what C does?

In theory, yes. It has every tool C has, and nothing stops it running the same checks. What it doesn't have is the rule to check the design first, or permission to stop. The results show that without those, it doesn't do it by itself.

Won't a better model make the procedure unnecessary?

I don't know yet, and I think it's a fair question. B's problem wasn't a lack of ability. It had the tools. The problem was that it never checked the design before answering, and nothing forced it to. A stronger model might check more often on its own, but that's something I'd need to test, not assume.

How do you know the result isn't just specific to your benchmark?

I can't fully rule that out. It's 64 cases I designed, plus four public datasets. Keeping 48 cases hidden means the system wasn't tuned to them, but the kinds of cases still reflect my choices. Testing on a broader, independent set of problems would be the way to show it generalises.

What would you do differently if you started again?

I'd plan for a bigger held-out set from the start, so the C versus A comparison had enough power. I'd get a second person reviewing the public-data labels from the beginning. And I'd test the scoring rubric with a small pilot before running the full blind round, which would have caught the problem early.

How I would develop this project

in the order I would do it

1. Settle the C over A question

This is the first thing I'd do. At the size of effect I saw, my test only had about a 45% chance of detecting it, which is basically a coin flip. To get an 80% chance, I'd need around 90 held-out cases, roughly double what I have now. That number assumes the real effect is as big as the one I measured, which tends to be optimistic, so I'd treat 90 as a minimum.

2. Ask the user for the design card

Right now the design card comes with each benchmark case. In a real tool, nobody hands you that. So the next version should ask the user short, plain questions before it picks a method. Were the same people measured more than once? Are these readings in time order? That's the biggest step between this being a research system and something people can actually use.

3. Test it on bigger models

I want to know whether the procedure still helps as models get more capable, or whether a strong enough model starts checking the design on its own. Because everything is frozen and repeatable, I can run the exact same evaluation on a larger model and compare directly.

4. Cover more kinds of study design

At the moment the system can only decline for designs like repeated measures, mixed models, time series and survival data. Adding those methods would let it actually analyse them instead. The bike sharing case, for example, could get a proper time-series analysis rather than just a refusal.

5. Fix the interpretation scoring

I'd run a second blind round with the rebuilt rubric, after a small pilot to make sure two scorers agree on what each level means. If they agree, I can finally report how good each system's written explanations are, which is the one metric I had to leave out.

6. Get an independent reviewer

Having a second person check the public-data labels would make the answer key stronger. Most labels come from how the data was built, but the 16 public ones would benefit from a second opinion.

7. Measure the effort and cost trade-off

I'd run each system at different effort levels to see how much accuracy you give up by making it faster or cheaper. That would show whether the procedure is worth it in a real setting where cost matters.

8. Turn it into a tool people can use

The end goal is something where you upload a dataset, answer a few questions about how it was collected, and get back a report you can trust, with every number traceable to a real calculation. It could plug into things people already use, like notebooks or spreadsheets. It could also work as a teaching tool, explaining to students why a method was chosen or rejected.